# fMRI prediction models — SOTA review

2026-06-14

## 1. Brain-JEPA

| | |
|---|---|
| **Status** | Published — NeurIPS 2024 Spotlight |
| **arXiv** | 2409.19407 |

### Architecture

JEPA (Joint Embedding Predictive Architecture) + ViT on **ROI time-series** (450 ROI × 160 timesteps). The predictor reconstructs the **embeddings** of target blocks instead of raw voxels (LeCun's I-JEPA paradigm adapted to fMRI).

### Pretraining dataset

- **UK Biobank** — 32,130 subjects (80% of 40,162 available), TR 0.735s

### Downstream evaluation datasets

- UK Biobank held-out (Sex)
- HCP-Aging (Sex)
- ADNI (n=189) — NC vs MCI, Amyloid β+/-
- MACC — NC vs MCI Asian cohort
- OASIS-3 — AD Conversion
- CamCAN — Depression

### Results (Accuracy / F1, %)

| Dataset | Task | Acc | F1 |
|---|---|---|---|
| UK Biobank | Sex | **88.17** | **88.58** |
| HCP-Aging | Sex | **81.52** | **84.26** |
| ADNI (n=189) | NC vs MCI | **76.84** | **86.32** |
| ADNI | Amyloid β+/- | 71.00 | 75.97 |
| MACC | NC vs MCI | 65.98 | 64.67 |
| OASIS-3 | AD Conversion | 69.00 | 67.32 |
| CamCAN | Depression | 72.73 | 67.45 |

### Protocol

Fixed 6:2:2 train/val/test split across all downstream datasets. Results averaged over 5 independent runs. Subject-awareness not explicitly stated.

## 2. BrainLM

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|---|---|
| **Status** | Published — ICLR 2024 Spotlight |
| **arXiv** | biorxiv 2023.09.12.557460 |

### Architecture

Vision Transformer as Masked Autoencoder (MAE). Masks 75% of input patches, reconstructs **raw values**. 111M parameters.

### Pretraining datasets

- **UK Biobank** — 76,296 recordings (~6,450 hours)
- **HCP** — 1,002 recordings (~250 hours)
- **Total** — 77,298 recordings / 6,700 hours
- Atlas: AAL-424 ROI

### Downstream evaluation datasets

- UK Biobank held-out only — 4 regression tasks

### Results (MSE on z-scored targets)

| Dataset | Task | MSE (z-scored) |
|---|---|---|
| UK Biobank | Age | **0.503 ± 0.021** |
| UK Biobank | PTSD (PCL-5) | 0.015 ± 0.0003 |
| UK Biobank | Anxiety (GAD-7) | 0.073 ± 0.003 |
| UK Biobank | Neuroticism | 0.069 ± 0.004 |

### Protocol

No Sex / MCI / AD / MMSE numbers reported. Only regression on UK Biobank held-out subjects.

# 3. SLIM-Brain

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|---|---|
| **Status** | Not published — preprint, under OpenReview review |
| **arXiv** | 2512.21881 |

### Architecture

**4D Hiera-JEPA** — Hiera (hierarchical pyramidal ViT, multi-stage) combined with JEPA paradigm. Input = **4D volumes** `(T, X, Y, Z)`. 4D tubelets, global attention within each stage.

### Pretraining datasets

- **HCP** (1 session per subject)
- **CHCP** (Chinese HCP, Ge 2023)
- **AOMIC PIOP1**
- **AOMIC PIOP2**
- **ABCD**
- **Total** — 4,129 sessions (70% of combined data)
- Harmonized to 2mm isotropic / TR 0.72s

### Downstream evaluation datasets

- HCP (Sex, Fingerprint) — internal
- ADNI (MCI classification)
- ADHD-200
- PPMI
- ABIDE (Age classification + regression)

### Results

| Dataset | Task | Acc | F1 / other |
|---|---|---|---|
| HCP | Sex | **91.1** | F1 **91.1** |
| HCP | Fingerprint | 98.5 | F1 98.1 |
| ADNI | MCI classification | **69.12** | F1 68.96 |
| ADHD-200 | ADHD | 63.53 | — |
| PPMI | Parkinson | 70.40 | — |
| ABIDE | Age (classification) | 64.41 | — |
| ABIDE | Age (regression) | — | MSE 0.2175 |

### Protocol

70/10/20 train/val/test split, 3 independent runs. Subject-awareness not explicitly stated.

# 4. BrainGFM

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|---|---|
| **Status** | Not published — preprint |
| **arXiv** | 2506.02044 |

### Architecture

**Graph Foundation Model** on connectomes. Combines **Graph Contrastive Learning + Graph Masked Autoencoder** + meta-learning + graph prompts + language prompts. Multi-atlas (8 parcellations).

### Pretraining datasets

- **27 datasets** covering 25 disorders
- **25,000+ subjects / 60,000 scans / 400,000 graph samples**
- Identified examples: OpenNeuro, UK Biobank, HCP, ABIDE, ABIDE II, ADHD-200, OASIS, ADNI 2, HBN, SubMex_CUD, UCLA_CNP
- Full list in Appendix N of the paper

### Downstream evaluation datasets

- ADHD-200 (ADHD)
- ABIDE II (Autism)
- **ADNI 2 (Alzheimer)**
- HBN (Depression, Anxiety, OCD, PTSD)
- SubMex_CUD (Cocaine Use Disorder)
- UCLA_CNP (Schizophrenia, Bipolar)

### Results (AUC / Accuracy, %)

| Dataset | Task | AUC | Acc |
|---|---|---|---|
| ADHD-200 | ADHD | 70.6 | 72.2 |
| ABIDE II | Autism | 71.2 | 73.5 |
| **ADNI 2** | **Alzheimer (AD)** | **80.3** | **85.1** |
| HBN | Major Depression | 83.6 | 85.5 |
| HBN | Anxiety | 85.2 | 86.3 |
| HBN | OCD | 80.4 | 85.8 |
| HBN | PTSD | 83.2 | 86.3 |
| SubMex_CUD | Cocaine | 71.1 | 74.6 |
| UCLA_CNP | Schizophrenia | 84.2 | 86.7 |
| UCLA_CNP | Bipolar | 73.5 | 76.3 |

### Protocol

Schaefer-100 atlas used in main results. No Sex / Age / Parkinson reported.

## 5. LCM (Large Connectome Model)

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|---|---|
| **Status** | Published — AAAI 2026 |
| **arXiv** | 2510.18910 |

### Architecture

**Decoder-only Transformer** (GPT-style) on connectomes (FC matrix). MHSA + MHCA between connectome features and “Brain-Environment-Interaction” tokens (encoding covariates: sex, age, scanner, site).

### Pretraining datasets

| Dataset | Subjects | Scans |
|---|---|---|
| HCP-Aging | 713 | 4,863 |
| HCP-YA | 248 | 3,293 |
| ADNI | 138 | 138 |
| PPMI | 209 | 209 |
| ABIDE | 1,025 | 1,025 |
| Taowu | 40 | 40 |
| Neurocon | 41 | 41 |
| **Total** | — | **~10,036** |

### Downstream evaluation datasets

- HCP-Aging (Sex)
- HCP-YA (Sex)
- ABIDE (Sex, Autism)
- ADNI (Alzheimer)
- PPMI (Parkinson)

### Results (F1, %)

| Dataset | Task | F1 |
|---|---|---|
| HCP-Aging | Sex | **73.94 ± 2.45** |
| HCP-YA | Sex | **72.23 ± 1.92** |
| ABIDE | Sex | 87.34 ± 4.48 |
| **ADNI** | **Alzheimer** | **85.33 ± 7.35** |
| PPMI | Parkinson | 84.18 ± 11.63 |
| ABIDE | Autism | 72.50 ± 1.91 |

### Protocol

Subject-aware 5-fold CV for HCPA/HCPYA/ADNI, 10-fold for others. Pretraining and finetuning always from the same CV fold’s training set to prevent leakage.

# 6. BNT (Brain Network Transformer)

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|---|---|
| **Status** | Published — NeurIPS 2022 |
| **arXiv** | 2210.06681 |

### Architecture

Pure Transformer on connectome graphs (not a GNN). Each ROI = one token, node features = full connection profile. Key innovation: **Orthonormal Clustering Readout** — pooling via K learned orthogonal prototypes.

### Pretraining

None — supervised end-to-end training.

### Downstream evaluation datasets

- ABIDE (Autism)
- ABCD (Sex, n=7,901)

### Results

| Dataset | Task | Metric | Score |
|---|---|---|---|
| ABIDE | Autism | AUROC | **80.2%** |
| ABCD | Sex (n=7,901) | — | not extracted |
| ADNI | NC vs MCI | Acc | **78.90** |

### Protocol

Subject-level splits. ADNI NC/MCI Acc reported as comparator within Brain-JEPA paper (Brain-JEPA Acc 76.84%, so BNT beats Brain-JEPA on this task).

## 7. OViTAD

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|---|---|
| **Status** | Published — Brain Sciences 2023 (MDPI journal, not a top venue) |
| **arXiv** | biorxiv 2021.11.27.470184 |

### Architecture

Standard **ViT-B/16** trained on individual **2D fMRI slices** (no 3D, no temporal modeling). Subject-level inference via **majority vote** across slices. No architectural innovation — only hyperparameter tuning.

### Pretraining

None — supervised end-to-end training.

### Downstream evaluation dataset

- ADNI rs-fMRI — 284 subjects total
- Split: 226 train / 27 val / **31 test** (80/10/10 at participant level)

### Results (F1)

| Dataset | Task | F1 |
|---|---|---|
| ADNI (n=284) | AD vs HC | **0.99 ± 0.02** |
| ADNI (n=284) | HC vs MCI | **0.97 ± 0.03** |

### Protocol

Test set = only 31 subjects. σ=0.02 on n=31 is noise-dominated — interpret with caution.

## 8. BrainNetCNN

| | |
|---|---|
| **Status** | Published — NeuroImage 2017 |
| **arXiv** | Kawahara 2017 PDF |

### Architecture

CNN with **3 custom topology-aware filters** for connectivity matrices:

- **Edge-to-Edge (E2E)** — aggregates row + column per cell
- **Edge-to-Node (E2N)** — aggregates per ROI
- **Node-to-Graph (N2G)** — global pooling

### Pretraining

None — supervised end-to-end training.

### Downstream evaluation dataset (original paper)

- Preterm infant DTI (diffusion MRI, **not fMRI**)
- Task: Bayley-III developmental scores

### Results

| Dataset | Task | Score |
|---|---|---|
| Preterm DTI (orig paper) | Bayley-III scores | **not comparable** |

### Protocol

Original architecture is now widely reused as a supervised baseline on adult fMRI by BNT / BrainGFM / BrainGB.

# 9. SwiFT

| | |
|---|---|
| **Status** | Published — NeurIPS 2023 |
| **arXiv** | 2307.05916 |

### Architecture

**4D Swin Transformer** on raw fMRI volumes (no atlas). 4D windows `(X, Y, Z, T)` + windowed self-attention + shifted windows between layers + hierarchical (4 stages). Closest volumetric Transformer paradigm to ours.

### Pretraining

Light contrastive SSL (composition not specified in main paper).

### Downstream evaluation datasets

- HCP-YA (Sex, Age, Cognitive intelligence)
- ABCD (Sex, Age, Intelligence)
- UK Biobank (Sex, Age)

### Results (extracted from comparator tables in Brain-JEPA & SLIM-Brain)

| Dataset | Task | Acc | Other |
|---|---|---|---|
| OASIS-3 | AD Conversion | 65.00 | F1 66.80 |
| ADNI | MCI | 64.45 | — |
| ABIDE | Age (cls) | 62.22 | MSE 0.4137 |
| ADHD-200 | ADHD | 60.81 | — |
| PPMI | Parkinson | 58.10 | — |
| HCP / ABCD / UKB | Sex, Age | — | not extracted |

### Protocol

Numbers reported as comparator within Brain-JEPA Table 9 and SLIM-Brain Table 1. Original SwiFT-paper numbers on HCP/ABCD/UKB not extracted in this pass.

## Datasets — dimensions and access

| Dataset | # subj. | TR (s) | Typical T | Access mode |
|---|---|---|---|---|
| UK Biobank | 40,000+ | 0.735 | 490 | Paid DUA (~£500-1000), 2-6 months |
| HCP-YA | 1,200 | 0.72 | 1,200 | Open with DUA, fast |
| HCP-Aging | 700 | 0.8 | ~470 | NDA controlled, 1-3 months |
| CHCP | ~370 | 0.72 | varies | Chinese open access |
| AOMIC PIOP1 | 216 | 2.0 | ~480 | Open |
| AOMIC PIOP2 | 226 | 2.0 | ~480 | Open |
| ABCD | 11,800+ | 0.8 | ~375 | NDA controlled, 1-3 months |
| ADNI / ADNI 2 | ~1,000 | 3.0 | ~140 | DUA, 1-2 weeks |
| ABIDE I | 1,112 | 1.5-3 | 76-300 | Open (NITRC / PCP) |
| ABIDE II | 1,114 | varies | varies | Open |
| ADHD-200 | 776 | 1.5-2.5 | 70-280 | Open (NITRC) |
| PPMI | ~500 | 2.4 | ~200 | DUA, 1-2 weeks |
| HBN | 3,000+ | 0.8 | varies | Open (AWS S3) |
| OASIS-3 | 1,098 | 2.2 | ~180 | Registration + proposal, 1-7 days |
| MACC | restricted | varies | varies | Restricted (Asian collaborators) |
| CamCAN | 700 | 1.97 | 261 | Open with registration |
| SubMex_CUD | restricted | varies | varies | Restricted |
| UCLA_CNP | 272 | 2.0 | 152 | Open (OpenNeuro) |
| Taowu | 40 | 2.0 | ~120 | Open (OpenNeuro) |
| Neurocon | 41 | 2.0 | ~140 | Open (OpenNeuro) |

**Access categories (in order of effort)**

- **Open access** (immediate via Python libs like `nilearn`): ABIDE I/II, ADHD-200, AOMIC PIOP1/PIOP2, CHCP, HBN, UCLA_CNP, Taowu, Neurocon
- **Registration + short proposal** (1-7 days): OASIS-3, CamCAN
- **DUA, fast** (1-2 weeks): ADNI, PPMI, HCP-YA
- **NDA controlled** (1-3 months): HCP-Aging, ABCD
- **Paid DUA, slow** (2-6 months): UK Biobank (~£500-1000)
- **Restricted to specific collaborators**: MACC, SubMex_CUD

## Summary tables

### Pretraining datasets vs evaluation datasets

| Model | Pretraining datasets | Evaluation datasets |
|---|---|---|
| Brain-JEPA | UK Biobank | UKB, HCP-Aging, ADNI, MACC, OASIS-3, CamCAN |
| BrainLM | UK Biobank + HCP | UK Biobank only |
| SLIM-Brain | HCP + CHCP + AOMIC PIOP1/PIOP2 + ABCD | HCP, ADNI, ADHD-200, PPMI, ABIDE |
| BrainGFM | 27 datasets (HCP, UKB, ABIDE, ADHD, OASIS, ADNI 2, HBN, …) | ADNI 2, ABIDE II, ADHD-200, HBN, SubMex_CUD, UCLA_CNP |
| LCM | HCPA + HCPYA + ADNI + PPMI + ABIDE + Taowu + Neurocon | HCPA, HCPYA, ABIDE, ADNI, PPMI |
| BNT | None (supervised end-to-end) | ABIDE, ABCD |
| OViTAD | None (supervised end-to-end) | ADNI (284 subjects) |
| BrainNetCNN | None (supervised end-to-end) | Preterm DTI (original paper) |
| SwiFT | Light contrastive SSL (corpus unspecified) | HCP, ABCD, UK Biobank |

### Labels used per model

| Model | Labels reported |
|---|---|
| Brain-JEPA | DX (NL/MCI), Amyloid β +/-, Sex, Depression |
| BrainLM | Age, PTSD, Anxiety, Neuroticism (z-scored MSE regression) |
| SLIM-Brain | DX (MCI), Sex, Fingerprint, ADHD, Parkinson, Age |
| BrainGFM | ADHD, ASD, AD (DX), MDD, Anxiety, OCD, PTSD, Cocaine, Schizo, Bipolar (10 disorders) |
| LCM | Sex, DX (AD, PD, ASD, SZ) |
| BNT | DX (Autism), Sex |
| OViTAD | DX (AD vs HC, HC vs MCI) |
| BrainNetCNN | Bayley-III scores (preterm) |
| SwiFT | Sex, Age, Cognitive intelligence |

## Notes on comparability

- **Most SOTA papers report Accuracy (%) and F1 (%)**, not AUC. Direct AUC-to-Acc conversion is approximate.
- **Brain-JEPA / BrainGFM / LCM use DX clinical diagnosis labels** (NL / MCI / AD from ADNI metadata). Direct comparison to CDR-based binarisations requires label alignment.
- **No SOTA paper reports horizon-specific cognitive decline forecasting** (1Y / 2Y / 3Y splits). Closest comparator: Brain-JEPA OASIS-3 AD Conversion (single horizon, Acc 69.00 / F1 67.32).
- **Subject-level splits**: Brain-JEPA (6:2:2), SLIM-Brain (70:10:20), LCM (subject-aware 5/10-fold), OViTAD (80:10:10 at participant level). Subject-awareness explicitly confirmed only for LCM and OViTAD.

## Sources

1. **Brain-JEPA: Brain Dynamics Foundation Model with Gradient Positioning and Spatiotemporal Masking** — Dong & Li et al., NeurIPS 2024 Spotlight — arxiv.org/abs/2409.19407
2. **BrainLM: A foundation model for brain activity recordings** — Ortega Caro et al., ICLR 2024 — biorxiv 10.1101/2023.09.12.557460
3. **SLIM-Brain: A Data- and Training-Efficient Foundation Model for fMRI Data Analysis** — Anonymous, arXiv Dec 2025 — arxiv.org/abs/2512.21881
4. **Brain Graph Foundation Model (BrainGFM)** — arXiv 2025 — arxiv.org/abs/2506.02044
5. **Large Connectome Model (LCM)** — AAAI 2026 — arxiv.org/abs/2510.18910
6. **Brain Network Transformer (BNT)** — Kan, Cui et al., NeurIPS 2022 — arxiv.org/abs/2210.06681
7. **OViTAD** — Sarraf et al., Brain Sciences 2023 — biorxiv 10.1101/2021.11.27.470184
8. **BrainNetCNN** — Kawahara et al., NeuroImage 2017 — PDF
9. **SwiFT** — Kim et al., NeurIPS 2023 — arxiv.org/abs/2307.05916